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| Name: | Class: | Class Register Number: |
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**中正中學**

**CHUNG CHENG HIGH SCHOOL (MAIN)**

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**PRELIMINARY EXAMINATION 2026  
SECONDARY 4**

**ADDITIONAL MATHEMATICS**

**4049/02**

**Paper 2**

**Wednesday 26 August 2026**

Candidates answer on the Question Paper.

**2 hours 15 minutes**

**READ THESE INSTRUCTIONS FIRST**

Write your name, class and index number in the spaces at the top of this page.  
 Write in dark blue or black pen.  
 You may use an HB pencil for any diagrams or graphs.  
 Do not use paper clips, glue or correction fluid.

Answer **all** the questions.  
 Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.  
 The use of an approved scientific calculator is expected, where appropriate.  
 You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.  
 The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 90.

| For Examiner's Use |                |
|--------------------|----------------|
| Question Number    | Marks Obtained |
| 1                  |                |
| 2                  |                |
| 3                  |                |
| 4                  |                |
| 5                  |                |
| 6                  |                |
| 7                  |                |
| 8                  |                |
| 9                  |                |
| 10                 |                |
| 11                 |                |
| Total Marks        |                |

This document consists of **20** printed pages.

**Mathematical Formulae****1. ALGEBRA***Quadratic Equation*

For the equation  $ax^2 + bx + c = 0$ ,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

*Binomial expansion*

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where  $n$  is a positive integer and  $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1) \dots (n-r+1)}{r!}$ .

**2. TRIGONOMETRY***Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

*Formulae for  $\triangle ABC$* 

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} ab \sin C$$

- 1 Scientists monitoring a shrinking glacier estimate that its area,  $A$  km<sup>2</sup>, is modelled by  $A = 850e^{-kt}$ , where  $t$  is the number of years after the year 2010 and  $k$  is a positive constant. In 2025, the area of the glacier was 620 km<sup>2</sup>.

(a) Find the value of  $k$ , correct to 4 significant figures. [3]

(b) A journalist reported: "*At this rate, the glacier would have completely disappeared by the year 2110.*" Do you agree with the journalist's statement? Use calculations to justify your answer. [2]

(c) Give a mathematical reason why the model may not be reliable to determine if the glacier will completely disappear. [1]

- 2 (a) Show that the equation  $(a^2 - bc)x^2 + 2(b + c)x = 4$  has real roots for all real values of  $a$ ,  $b$  and  $c$ . [4]

- (b) Find the range of values for the constant  $h$  for which the curve  $y = x^2 - 3hx + 7 - h$  lies above the line  $y = 1 - hx$ . [4]

- 3 (a) Solve the equation  $\log_3(1+2x) + 2\log_9(3x-1) = \log_6 1$ .

[5]

- (b) Solve the equation  $5^x - 4(5^{-x}) = 3$ , leaving your answer correct to 2 significant figures. [5]

4     (a) A curve has equation  $y = \ln\left(\frac{3x+2}{\sqrt{x+5}}\right)$ , where  $x > 0$ .

(i) Find the gradient of the curve at the point where  $x = 1$ . [3]

(ii) Explain why the curve is strictly increasing for all  $x > 0$ . [2]

- (b) The curve  $y = f(x)$  is such that  $f'(x) = 4 \sin 2x \cos 2x$ . The curve passes through the point  $\left(\frac{\pi}{8}, 1\right)$ . Find an expression for  $f(x)$ . [4]

5 A curve has equation  $y = 2e^x + 3e^{-2x}$ .

- (a) Find  $\frac{dy}{dx}$  and hence show that the  $x$ -coordinate of the stationary point satisfies the equation  $e^{3x} = 3$ . [3]

- (b) Without solving for the value of  $x$  at the stationary point, determine the nature of the stationary point. [3]

- 6 The polynomial  $f$  is given by  $f(x) = x^3 + (k+1)x^2 + 3kx + 2k$ , where  $k$  is constant.  
It is also given that  $f(x) = (x+1)(x^2 + px + q)$ , where  $p$  and  $q$  are constants.

(a) Express  $p$  and  $q$  in terms of  $k$ .

[3]

(b) Find the values of  $k$  for which  $f(x) = 0$  has exactly two real roots.

[3]

7 A circle, with centre  $C$ , has an equation  $x^2 + y^2 - 8x - 6y + 12 = 0$ .

- (a) Find the coordinates of  $C$  and the radius of the circle. [3]

The point  $R(6, 6)$  lies on the circle.

- (b) Verify that  $R$  lies on the line  $2x + 3y = 30$  and show that this line is tangent to the circle at  $R$ . [3]

The tangent at  $R$  meets the  $x$ -axis at  $A$ .

(c) Find the area of triangle  $ACR$ .

[3]

- 8** A particle moves in a straight line and passes through a fixed point  $O$  with a velocity of 10 m/s. The acceleration,  $a$  m/s<sup>2</sup>, of the particle  $t$  seconds after passing  $O$ , is given by  $a = -3(t+2)^2 + 9$ .
- (a)** Explain, by showing your working clearly, why the particle only comes to instantaneous rest once in the interval  $0 \leq t \leq 3$ . [6]

- (b) Find the total distance travelled by the particle in the interval  $0 \leq t \leq 3$ . [4]

- 9 (a) Prove the identity  $(\cot A - \tan A)\sin A = 2\cos A - \sec A$ . [4]

- (b) Solve the equation  $(\cot x - \tan x)\sin x = 1$  for  $0 \leq x \leq 2\pi$ . [4]

- (c) The equation  $2 \cos kx - \sec kx = -\frac{7}{2}$  has exactly four solutions for  $0 \leq x \leq 2\pi$ , where  $k$  is a positive integer. Find the value of  $k$ . [4]

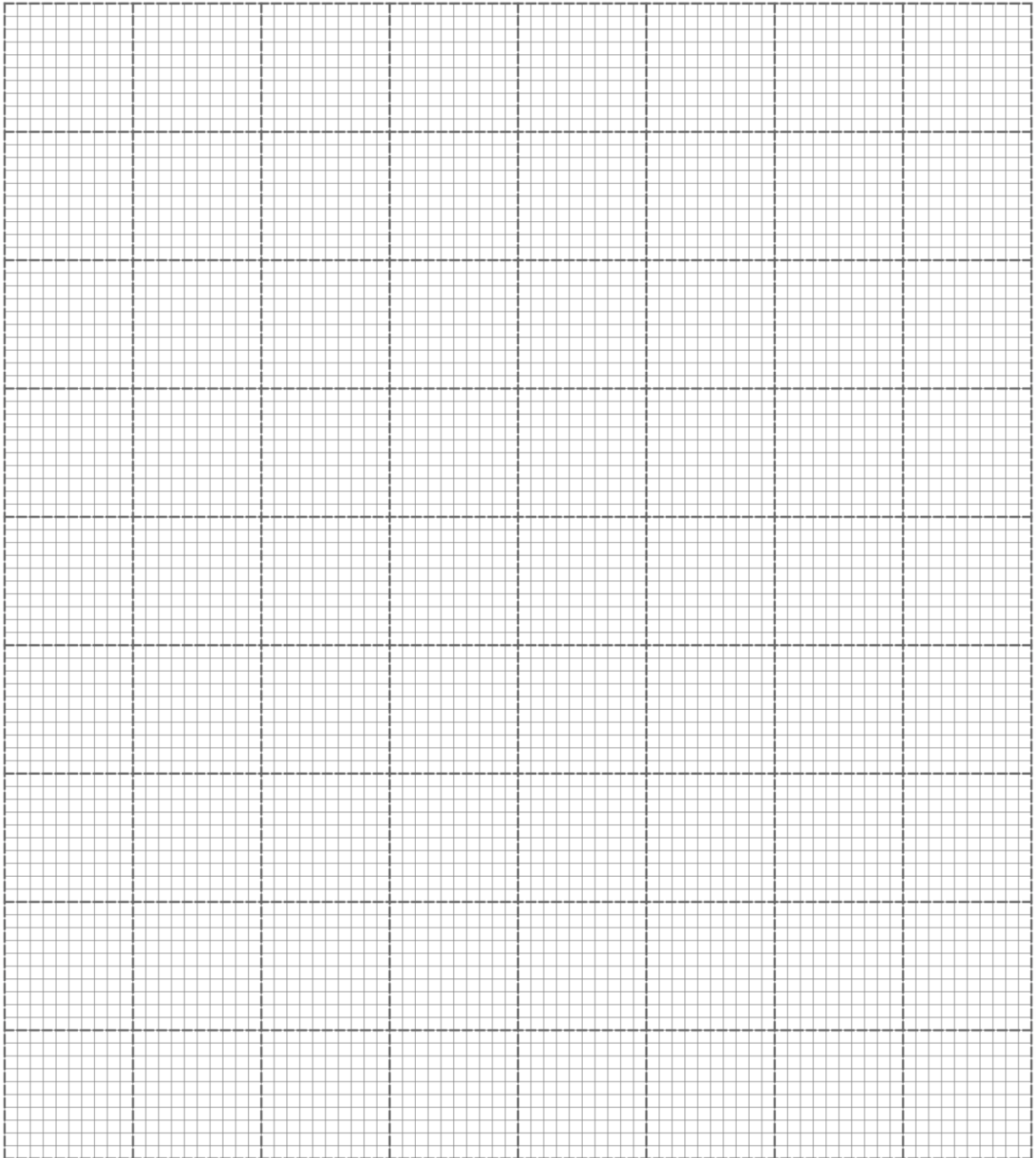
- 10 The table shows the experimental values of two variables  $x$  and  $y$ .

|     |      |      |      |      |     |
|-----|------|------|------|------|-----|
| $x$ | 1    | 1.5  | 2    | 2.5  | 3   |
| $y$ | 10.2 | 13.4 | 14.1 | 11.7 | 5.8 |

It is known that  $x$  and  $y$  are related by the equation  $y = px^3 + qx$ , where  $p$  and  $q$  are constants.

- (a) Plot  $\frac{y}{x}$  against  $x^2$  and draw a straight line graph.

[3]



(b) Use your graph to estimate the value of  $p$  and of  $q$ .

[3]

(c) Another straight line can also be drawn from the equation  $y = px^3 + qx$ .  
Find the gradient and vertical intercept of this line.

[2]

- 11 (a)** Differentiate  $\frac{x}{(1+x)^2}$  with respect to  $x$ . [2]

- (b)** Hence, evaluate  $\int_0^1 \frac{3-x}{(1+x)^3} \, dx$ . [4]